

Counterfactual vs. Prefactual: Two narrative AIs improve causability for health data by different mechanisms

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Abstract

Narrative artificial intelligence (AI), which adds narrative persuasion to simple causal interpretation, is an explainable AI technique that helps patients understand their health data (e.g., MRI results). Narrative AI is expected to improve causability for health data and thereby induce meaningful behavioral change. However, research on which type of narrative AI (i.e., counterfactual or prefactual) better improves causability for complex health data is still elusive. This study created two different types of narrative AI for MRI reports (i.e., counterfactual and prefactual) and compared their impact on causability with 20 participants. System causability scale results showed that both narrative AI provided a high level of causability. Semi-structured interview results showed that the two narrative AIs improve causability by different mechanisms. While counterfactual made participants focus on self-reflection, prefactual made participants seek self-improvement to prepare for the future. Our findings showed that the two types of narrative AI (i.e., counterfactual and prefactual) improve causability by different mechanisms. By understanding these different underlying mechanisms, we expect to be able to design and deliver more personalized narrative AI to users.

Keyword

Narrative AI; Counterfactual; Prefactual; Causability; Health data.

1. Introduction

Brain magnetic resonance imaging (MRI) is an important health data that can be used to detect

Alzheimer's disease (AD) through structural analysis of the brain, such as the quantification of brain atrophy [1]. Neurologists interpret these MRI results for patients, but it is difficult for the patients to fully understand the results because the information is usually communicated in technical terms over a short period of time [2]. Most patients do not have medical knowledge of brain structures, so it is difficult for them to understand MRI results on their own [3]. Explainable artificial intelligence (XAI) techniques, which provide interpretable explanations for complex results, can help patients understand MRI results at any time without the need for specialized knowledge [4]. In this context, the demand for XAI research for health data is increasing [5].

Recent XAI techniques can provide patients with interpretations of MRI results in various ways. For example, IBM Watson OpenScale is an AI platform that helps users understand MRI results by exploring brain volume and demographic information [6]. A class activation map helps users visually identify structural abnormalities in the brain through heatmaps [7]. Although these XAI techniques can provide some degree of interpretation, they are limited to explaining 'causality' (i.e., a simple causal interpretation) [8]. This simple causal interpretation is not enough to lead to meaningful changes in user behavior. To genuinely change user behavior, Holzinger and colleagues emphasized the need for 'causability,' a causal interpretation with effectiveness, efficiency, and satisfaction in a specified context of use [9]. How to improve causability for complex health data such as MRI results is still an open question [10].

In recent studies [10, 11], it is expected that 'narrative AI,' which adds narrative persuasion to causal interpretation, can be used as a new XAI technique to improve causability. Narrative AI can increase causability by allowing users to meaningfully interpret complex results and play a central role in decision-making with AI [12]. As research has progressed, two types of narrative AI have emerged [13]. First, 'counterfactual,' which focuses on the interpretation of the past, is useful for making users look back and reflect on themselves [13]. In contrast, 'prefactual,' which places more importance on the interpretation of the future, is more suitable for inducing users to take actions that prevent negative consequences in the future [14]. It is unclear which narrative AI is more suitable for improving patients' causability for health data.

In this study, we compared the impact of two different types of narrative AI (i.e., counterfactual vs. prefactual) on causability for health data (i.e., MRI results). To this end, Section 2 introduces narrative AI and its impact on causability. Section 3 describes the experimental procedure using narrative AI. Section 4 shows the impact of narrative AI on patients' causability on MRI results. Finally, Section 5 presents how narrative AI can be used to improve patients' causability for health data.

2. Two types of narrative AI: Counterfactual vs. Prefactual

Wadsley and Ryan [15] described 'narrative AI' as a novel XAI technique that can explain to users the meaning of causal relationships between inputs and outputs. Dai and colleagues [13] suggested two types of narrative AI according to the differences in narrative persuasion styles (i.e., counterfactual vs. prefactual). As shown in Figure 1, 'counterfactual' narrative AI explains how the present output would be different if the past inputs had been different [13]. Self-reflection of past behaviors in Alzheimer's disease patients is an example of counterfactual: "If I had worked

hard on cognitive enhancement activities in the past, I wouldn't have Alzheimer's now." [16] In contrast, 'prefactual' narrative AI simulates a change of possible future outputs based on the present input [17]. Prefactual like "Exercise every day, like now, and your cognitive function will improve in the future." can be used to persuade people to change their behavior [18, 19]. Even though both counterfactual and prefactual are useful XAI techniques for justifying the causal relationship between a model's inputs and outputs, it is not clear which one will be more helpful for improving user causability [10].

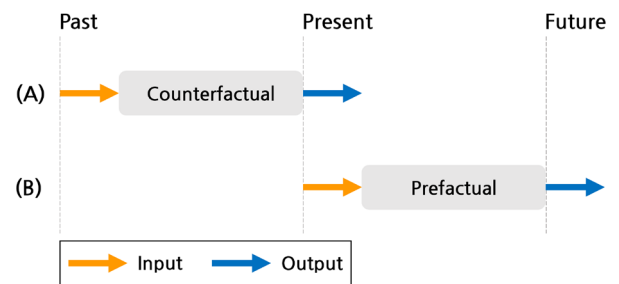


Figure 1 Two types of narrative AI. (A) Counterfactual describes how the present output would have been different if the past inputs were different. (B) Prefactual describes how future outputs can be realized differently depending on the present input.

In medicine, 'causability' refers to the explanatory power of inducing meaningful behavioral changes in patients [9]. Causability through XAI can increase the reliability of AI analysis results for medical images [20]. High causability helps patients to clearly understand their situation and context and make proactive decisions based on this [21]. As a result, XAI can support patients in taking a leading role in their medical decision-making process (e.g., taking medicine, receiving treatment, or changing behaviors) [12].

Counterfactual and prefactual improve causability by different mechanisms. Counterfactual makes people compare reality (i.e., the present situation due to past inputs) with conjecture (i.e., assumptions about the present situation that can occur when past inputs change) [13]. By comparing reality and conjecture, people can reflect on past actions

(i.e., inputs) that led to their present situation and prevent these past actions from repeating themselves [24]. In contrast, prefactual improves causability by providing conjectures (i.e., predictions of possible outcomes in the future based on the present input) [13]. By comparing different conjectures, people can recognize the different futures that lie ahead and decide their actions more proactively and confidently [23]. Cognitive psychology research shows that both counterfactual and prefactual lead to behavioral changes through emotional changes: counterfactual makes one feel guilty in the self-reflection process; prefactual motivates self-improvement by presenting various possibilities in the future [18, 25].

Although there have been several studies of both types of narrative AI, it is still unclear which one is more effective in improving causability. Gilbar and Hevroni studied the effects of counterfactual in cancer patients, but did not compare their effects to prefactual [16]. Bertolotti and his colleagues demonstrated that prefactual can be used to change the behavior of people, but they didn't compare the effects to counterfactual [18]. Dai and colleagues looked at how confident people were in counterfactual and prefactual when making decisions; however, this study didn't examine their impact on causability [13]. To our knowledge, there have been no studies comparing the effects of counterfactual and prefactual in terms of patients' causability for health data so far. In this context, this study compared which of the two types of narrative AI (i.e., counterfactual vs. prefactual) better improved patient causability for MRI results.

3. Materials and methods

3.1 Participants

We recruited 20 participants from the university community (average age is 22.45 ± 1.47 years; 50% female). Participants did not need to have expertise in AI, MRI, or the brain to participate in the experiment. Nine participants were familiar

with AI. Six participants had taken an MRI and seen doctors about it. The study was reviewed and approved by the Institutional Review Board and performed according to the tenets of the Declaration of Helsinki (HYUH-2021-08-020-004).

3.2 Two types of narrative AI for MRI report

As shown in Figure 2, we created two types of MRI reports (i.e., counterfactual and prefactual). MRI reports were divided into two parts: the AI analysis part and the narrative persuasion part. In the AI analysis part, the participants' scores for the hippocampus, ventricle, frontal lobe, entorhinal cortex, and memory function were equally displayed in the two types of reports as a radar chart. And we summed up the radar chart in one sentence: "Even though the size of the hippocampus and entorhinal cortex has shrunk, it seems to be keeping cognitive functions normal because it has a good memory."

There were differences in the title and explanation of counterfactual and prefactual in the narrative persuasion part. In the counterfactual MRI report, we named the title "The impact of past inputs on present outcomes" and used "If you did something, you would do something." as the conditional proposition. While, in the prefactual MRI report, we named the title "The impact of present inputs on future outcomes" and used "If you do something, you will do something." as the conditional proposition.

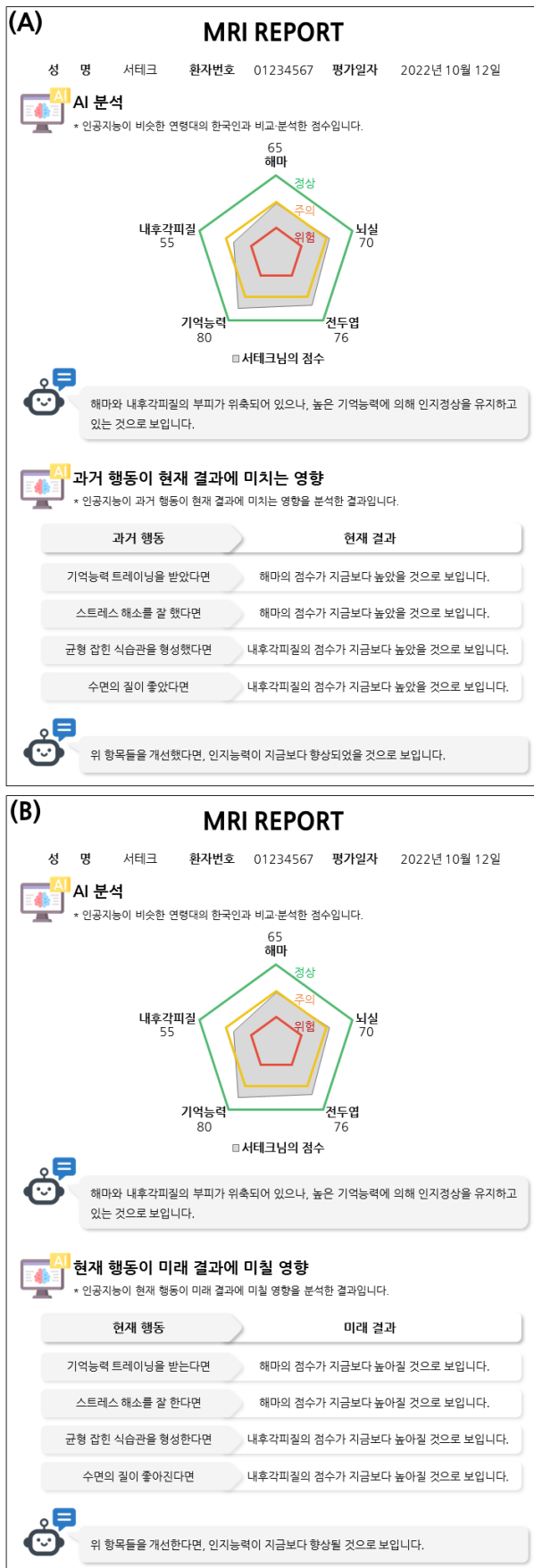


Figure 2 Two types of MRI report: (A) Counterfactual; (B) Prefactual.

3.3 System causability scale

System causability scale (SCS) is a questionnaire that assesses causability quality on a 5-point Likert scale, ranging from strongly disagree (1) to strongly agree (5) [27]. As shown in Table 1, we used seven questions to rate the improvement of causability for MRI reports through the narrative AI's explanation.

Table 1. System causability scale (SCS) questionnaire, adapted from [27].

Questions
1 I understood the MRI report within the context of my health.
2 I did not need support to understand the MRI report.
3 I found the MRI report helped me understand the causality.
4 I did not find inconsistencies in the MRI report.
5 I was able to use the MRI report with my knowledge base.
6 I think that most people would learn to understand the MRI report very quickly.
7 I did not need more references in the MRI report: e.g., medical guidelines and regulations.

3.4 Experimental procedure

We conducted experiments using a video conferencing platform (i.e., Zoom). Written informed consent was obtained after fully explaining the experiment to the participants. To evaluate the improvement of causability through narrative AI on MRI reports, participants were asked to read the following hypothetical scenario: "Recently, you often forget what you had to do or did. So you had a brain MRI and a working memory test at the hospital. The AI analysis report for brain MRI has arrived and you are about to read it before seeing a doctor."

After reading the scenario, participants were randomly assigned one of the two types of

narrative AI on MRI reports (i.e., counterfactual or prefactual). Participants could read the MRI report until they fully understood it. After reading an MRI report, participants rated SCS questionnaire (see Table 1). After the questionnaire, we conducted semi-structured interviews with participants as follows: “What do you think of the MRI report provided by narrative AI? What was the difficulty level of the MRI report? What did you think after reading the narrative persuasion part?” [26] The average duration of reading the MRI report was 1.58 ± 0.60 minutes and the average duration of the semi-structured interview was 6.44 ± 1.76 minutes. Participants could stop the experiment at any time they wanted, but no one stopped the experiment.

3.5 Data analysis

We used SPSS 27 to analyze the data. First, a χ^2 -test and a Mann-Whitney U test were used to examine whether there were differences in the basic demographic characteristics of participants (i.e., gender and age) who used each narrative AI. Second, a Mann-Whitney U test was used to compare SCS results between the two types of narrative AI (i.e., counterfactual and prefactual). Lastly, through the analysis of interview results, we explored how underlying mechanisms such as self-reflection and self-improvement occur in the two types of narrative AI.

4. Results

4.1 Basic demographic characteristics

A χ^2 -test showed that there was no significant difference in gender between participants groups ($\chi^2(1) = 0.000$, $p = 1.000$). A Mann-Whitney U test showed that there was no statistical difference in age between groups who read the counterfactual report and those who read the prefactual report ($p = 0.559$).

4.2 System causability scale results

SCS results between the two types of narrative AI (i.e., counterfactual and prefactual) were compared using a Mann-Whitney U test. As

shown in Table 2, there were no significant differences in SCS results between counterfactual and prefactual. Although there was no difference between the two types of narrative AI, both provided a high level of causability.

Table 2. System causability scale results from counterfactual and prefactual.

	Counterfactual	Prefactual	<i>p</i> value
1	4.30±0.68	4.10±0.74	0.534
2	3.80±1.23	3.90±0.99	0.937
3	3.90±1.10	4.10±0.99	0.684
4	4.20±0.79	3.60±1.51	0.475
5	4.30±0.95	4.20±0.79	0.622
6	3.70±1.06	4.00±0.94	0.477
7	4.20±1.14	3.60±1.51	0.330

4.3 Interview results

For analysis, each interview was audio recorded and transcribed. We summarized the interview results into two parts: 1) common perceptions of narrative AI and 2) different mechanisms in counterfactual and prefactual. Each quote outlines whether the response came from a participant who read the counterfactual report (“C”) or the prefactual report (“P”).

4.3.1 Common perceptions of narrative AI

There was a common perception of both types of narrative AI. First, both narrative AI made it easy for participants to understand complex health data by putting it in the context of their health. Participants said it was not difficult to interpret and understand the MRI report with background knowledge: “I thought it could be difficult because there were many medical terms, but it was easy to read because of the narrative persuasion part (C8 and P2).” Second, both narrative AI improved the quality of causability. Most participants (16 out of 20) said that they could understand causality between the symptoms they experienced and the MRI results.

In this regard, C9 and P5 replied that “The MRI report was well organized so that it was easy to understand, so I thought it was connected from cause to effect.” Third, both narrative AIs made participants feel emotionally relaxed and gave them the idea that they could play a leading role in making decisions about their treatment. C5 responded that “I usually feel worried until the diagnosis results come out, but I think the advantage of the narrative AI is to resolve this anxiety quickly.” and C10 said that “Knowing my overall condition through narrative AI and having seen a doctor, I think I can more easily understand the doctor’s diagnosis than before.” In sum, both types of narrative AI showed that they improved causability by providing complex health data easily and by allowing participants to play a central role in medical treatment.

4.3.2 Different mechanisms in counterfactual and prefactual

Depending on the type of narrative AI, participants focused on different parts of the information. Many participants in the counterfactual group (seven out of 10) asked for additional information about the AI analysis part (i.e., how to interpret the radar chart and the process by which the past results are analyzed). C1 said, “It would be nice if an explanation of the role or effect of each area was added to the chart.” On the other hand, most participants in the prefactual group (nine out of 10) demanded more specific information about the narrative persuasion part (i.e., what specific actions should be taken to change the future). P5 said, “It would be more satisfying if there were more information that could help understand the advice more clearly, such as a site or videos.” These findings showed that the information that the counterfactual group and the prefactual group consider important in information processing is different.

Depending on what information participants focused on, mechanisms for behavioral change also differed. Participants in the counterfactual group (four out of 10) responded that they regretted their past actions and reflected on why

these results came out. Then they said they would change their past behavior with the MRI report. C8 responded, “I got this result because I have lived in this pattern, so I need to look back on myself.” Whereas, most participants in the prefactual group (nine out of 10) replied that the past was already gone, so they thought about how to manage their health in the future, and they were motivated to change their behavior for a better future. P9 said that “I should change it from now on because I need to manage it well from now on to improve future results.” In sum, while the counterfactual group reached behavior change through self-reflection, the prefactual group reached behavior change for self-improvement.

5. Discussion and conclusion

In this study, we investigated the impact of two different types of narrative AI (i.e., counterfactual and prefactual) on causability for health data. SCS results showed that the two types of narrative AI provide similarly high levels of causability. The interview results showed that both types of narrative AI elicited behavioral changes in participants, but differed in the underlying mechanisms. Specifically, counterfactual made participants focus on the past, whereas prefactual made participants focus on the future.

Depending on whether participants focused on information from the past or the future, the mechanism of behavioral change was different. According to interviews, participants in the counterfactual group developed a flow of thoughts based on self-reflection. They tried to find the cause of the current results in their past behavior. For example, participants in the counterfactual group requested additional information about how to interpret radar charts in the AI analysis part or what role each area of the brain is related to. Meanwhile, participants in the prefactual group pursued self-improvement for a better future. They asked what actions could change their future. For example, the prefactual group asked for more

information about what actions can be done to improve memory, or about what balanced eating habits are, specifically in the narrative persuasion part. This shows that the interaction between humans and AI can be different depending on the type of narrative AI.

There are some limitations in this study. First, although a hypothetical scenario was provided so that participants could be immersed in the MRI report, the degree of immersion may vary depending on the individual experience of the participant. In future studies, utilizing MRI reports based on participants' actual health data will help to increase the validity of study findings. Second, since both MRI reports adopted the form of 'if A, then B' as the same form of a conditional proposition, participants may have perceived both types of MRI reports similarly. In the future, it would be desirable to generate narrative AI based on various types of conditional propositions using natural language processing technology and study in detail how participants perceive each conditional proposition differently.

Despite the aforementioned limitations, this study showed that counterfactual and prefactual could improve the quality of causability for complex health data. Furthermore, we showed that counterfactual participants base their thinking on self-reflection and prefactual participants focus their thinking on self-improvement. Both narrative AIs can be used by AI designers to design human-AI interactions for behavioral change.

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